

Request for Information

DARPA-SN-26-62

**Defense Advanced Research Projects Agency (DARPA)
Tactical Technology Office (TTO)**

Responses due June 1, 2026, by 4:00 PM Eastern Time (ET).

This Request for Information (RFI) from the Tactical Technology Office (TTO) within the Defense Advanced Research Projects Agency (DARPA) seeks information on graphene production and use in load-bearing structures for aerospace applications.

Responses to this RFI will be used to gather data for future possible programs.

Intellectual or other privileged or proprietary information contained in a response to this RFI will not be distributed outside of the U.S. Government. In the event that a new DARPA program is developed in response to this RFI and a solicitation is issued, no intellectual or other proprietary information received in response to this RFI will be divulged to entities outside of the U.S. Government.

BACKGROUND:

Graphene is a relatively new two-dimensional material with exceptional strength, high thermal and electrical conductivity, and very low weight. DARPA is interested in exploring the use of graphene as an ingredient in aerospace structures to increase strength, stiffness, and/or flexibility while reducing weight.

DESCRIPTION:

DARPA is interested in understanding the state-of-the-world and state-of-the-art for graphene production and manufacturing for potential use in load-bearing structures.

This RFI seeks information on:

1. The largest size (length, width, thickness) of graphene sheets available for use in structural parts today.
2. Potential solutions to scale up the size and manufacturability of graphene sheets for aerospace structures.
3. Use of graphene in the design and manufacturing of load-bearing structure—i.e., static and fatigue strength/stiffness (in plane compression/tension, out of plane and interlaminar strength, shear strength), part size, manufacturing techniques, etc. This can be as the main material, as an additive to other material or adhesives, or as a layer/ply in a composite part.
4. Test data that can substantiate the use of graphene as a structural material.
5. An estimate of weight reduction potential for graphene /graphene composite hybrid from traditional carbon composite/metallic structure.
6. An estimate of cost to produce aerostructures incorporating graphene. This can be expressed as dollar-per-pound (\$/lb), or some other measure as deemed appropriate.

REQUESTED INFORMATION:

Responses are welcome from all capable sources including, but not limited to, private or public companies, individuals, universities, University-Affiliated Research Centers (UARC)s, Federally Funded Research and Development Centers (FFRDCs), not-for-profit research institutions, U.S. Government-sponsored laboratories, large and small businesses, and nontraditional defense contractors.

DARPA requests responses **that address all** technical areas below:

1. Current capability to manufacture graphene sheets in large sizes compatible with aerospace load-bearing structures at any quantity.
2. Current capability to join together multiple graphene sheets to make larger structures.
3. Barriers to manufacturing graphene sheets in large sizes compatible with aerospace load-bearing structures at production-level quantity and quality.
4. Current capability to use graphene in manufacturing of load bearing structure. This can use graphene as the main material, as an additive to other material or adhesives, or as a layer/ply in a composite part.
5. An estimate of weight reduction for the same strength in a load bearing structure part.
6. An estimate of manufacturing costs for graphene aerostructures.

For the purposes of this RFI, DARPA is expressly **not interested** in the following:

1. Modeling or analysis techniques used for graphene material.
2. Theoretical assertions on the merits of graphene that are absent substantiating data.
3. Use of graphene to replace more orthodox methods of sensor construction and integration unless specifically tied to load bearing structures.
4. Use of carbon nanotube material or particles in these applications.

SUBMISSION FORMAT:

This announcement contains all information required to submit a response. No additional forms, kits, or other materials are needed.

General Guidelines:

1. Respondents to this RFI are encouraged to be as succinct as possible, while also providing actionable insight.
2. All submissions must be written in English.
3. White papers shall not exceed five (5) single-sided written pages using 11-point font with 1” margins all around. No smaller than 8-point font may be used for figures, tables, and charts.
4. Electronic submittal shall be in Microsoft Word, Microsoft PowerPoint, or PDF.
5. All documents submitted must be clearly labeled with the DARPA RFI number (DARPA-SN-26-62), respondent organization, and white paper title.
6. To the maximum extent possible, respondents are encouraged to send non-proprietary information.
 - o Respondents are responsible for clearly identifying proprietary information. Submissions containing proprietary information must have the cover page and each page containing

such information clearly marked with a label such as “Proprietary” or “Company Proprietary.” *NOTE: “Confidential” is a classification marking used to control the dissemination of U.S. Government National Security Information as dictated in Executive Order 13526 and should not be used to identify proprietary business information.*

7. Do not include elaborate brochures or marketing materials; only include information relevant to the areas identified under Requested Information.
8. Use of a diagram(s) or figure(s) to depict the essence of the response is permitted and counted towards page count.
9. Send unclassified submissions to DARPA-SN-26-62@darpa.mil. **DO NOT SUBMIT CLASSIFIED INFORMATION TO THIS ADDRESS.** It is not expected that the inclusion of higher classification information will be necessary for response to this RFI and submitters are specifically discouraged from including such information.
10. Submissions sent through other mediums, channels, or after the prescribed deadline will not be considered. Early responses are encouraged.

All white papers shall contain the following sections which are included in the page count:

- A. Cover Sheet. Provide the following information:
 - a. Response Title.
 - b. Technical point of contact name, organization, telephone number and email address (The contact information should include the respondent’s technical and/or administrative points of contact (names, addresses, phone numbers, fax numbers, and e-mail addresses) to enable potential clarification discussions.
 - c. Indicate the Key Challenge(s) addressed by this response.
 - d. Indicate classification level of response (e.g., Unclassified, Unclassified/CUI, Secret Collateral).
- B. Technical Description. **Address all** of the technical areas listed under Requested Information as stated in this RFI. Graphics within the technical descriptions are encouraged to the extent that they aid in succinctly describing the concepts.
- C. Graphic Overview Slide. If desired, include a single PowerPoint slide that graphically depicts the main ideas of the response. This will not count against the five (5) single-sided written pages.
- D. Bibliography/References.

All technical and administrative correspondence and questions regarding this announcement and how to respond to this RFI should be sent to DARPA-SN-26-62@darpa.mil. E-mails pertaining to this RFI that are sent directly to individual DARPA Program Managers will not receive a response. No telephone inquiries will be accepted.

NOTE: DARPA may conduct individual discussions with respondents as necessary to gain a full understanding of the technical and partnership models submitted. DARPA will contact respondents individually via e-mail.

ELIGIBILITY

DARPA invites white paper submissions from all those engaged in related research activities and appreciates responses from all capable and qualified sources, including, but not limited to, universities, UARCs, FFRDCs, private or public companies, and U.S. Government research laboratories, large and small businesses, and nontraditional defense contractors.

DISCLAIMERS AND IMPORTANT NOTES:

1. This RFI is NOT a Request for Proposals (RFP), a Program Solicitation (PS), nor a Request for Quotations (RFQ).
2. This is an RFI issued solely for market research information and new program planning purposes; it does not constitute a formal solicitation for proposals.
3. Responses received are not offers and will not be accepted by the Government to form a binding contract or agreement nor will responses bind DARPA to any further actions related to this topic, including requesting follow-on proposals from respondents to this RFI.
4. Responding to this RFI is voluntary and is not required to propose to a subsequent solicitation (if any) on this topic.
5. DARPA will not provide reimbursement for costs incurred in responding to this RFI. Respondents are solely responsible for all costs associated with responding to this RFI.
6. Respondents are advised that DARPA is under no obligation to acknowledge receipt of the response, information received or provide feedback to respondents with respect to any information submitted under this RFI.
7. DARPA will disclose submission contents only for the purpose of review. Submissions may be reviewed by the Government (DARPA and other Government partners), FFRDCs, and Scientific, Engineering and Technical Assistance (SETA) support contractors bound by appropriate non-disclosure agreements.
8. All information received in response to this RFI will be treated as confidential business information. Responders should properly mark their responses with a label such as “Proprietary” or “Company Proprietary” where applicable. All information received in response to this RFI that is marked “Proprietary” or “Company Proprietary” will be handled accordingly.
9. No questions concerning this RFI and the requirements for future solicitation(s) will be considered.

Point of Contact:

DARPA/TTO

DARPA-SN-26-62@darpa.mil